

MUZZAMMIL SARWAR

Senior Full-Stack Engineer & System Architect | TypeScript · Python · AI Systems
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SUMMARY

Senior full-stack engineer and system architect with 5 years across product platforms, AI systems and data tooling. Designs systems end to end, from data model and service boundaries to failure modes, and builds them quickly with AI-assisted development, keeping tests and code review as the quality gate. Currently sole engineer and architect of a multi-portal workforce platform with a cross-platform desktop agent. Previously built multi-agent AI systems, FastAPI backends, a production recommendation engine and n8n automations; a cross-platform test-automation framework and an Oracle-to-PostgreSQL migration tool; and MEAN-stack web applications. Strongest in TypeScript (NestJS, Next.js), Python (FastAPI), PostgreSQL and LLM integration. Open to remote roles.

SKILLS

Languages: TypeScript, JavaScript, Python, SQL, Bash, C#

Backend: Node.js, NestJS, Express, FastAPI, REST APIs, Microservices, BullMQ, Redis, Auth.js / JWT, Webhooks

Frontend: Next.js (App Router), React, Angular, Streamlit, Tailwind CSS, Radix UI, Zod

Data: PostgreSQL, MySQL, Prisma, pgvector, Oracle-to-PostgreSQL Migration, Schema Design, Query Optimisation

AI / LLM: OpenAI, Gemini (Vision), RAG & Embeddings, Multi-Agent Systems (CrewAI, LangChain), LightFM Recommenders

Automation & Desktop: n8n, Event-Driven Workflows, Electron, Native Modules & FFI, Windows / macOS / Linux Builds

DevOps & Testing: Docker, GitHub Actions, Caddy, Cloudflare, Railway, Linux, Jest, Python Test Automation

AI-Assisted Development: Claude Code, Cursor: architecture-first, agent-driven implementation with test and review gates

EXPERIENCE

Full-Stack Developer | Growth Rise Solutions · Remote

May 2026 – Present

Sole engineer and architect of Controla-VA, a workforce time-tracking and productivity platform for distributed teams

- Architected and built the platform end to end: five role-based Next.js portals on a NestJS API (30+ modules, PostgreSQL, Prisma) with office-scoped access control and a full audit trail.
- Built the Electron desktop agent running on ~50 workstations across Asia and Latin America (offline-first sync, multi-monitor screenshots, native C# URL capture, silent auto-updates); porting it to macOS and Linux.
- Root-caused an over-billing defect (a 17-minute session recorded as 16.5 hours) and fixed it server-side for every installed version; designed an event-sourced billing model now running in shadow mode.
- Shipped AI features to production: RAG candidate matching on pgvector, Gemini vision screenshot classification, and LLM invoice and CV parsing.
- Built a multi-currency finance platform that reconciles bank, PayPal, Payoneer, Airwallex and eBay statements into a three-tier P&L, with a grounded AI assistant.
- Integrated Zoom meetings with recording and AI summaries; run staging and production on Docker, Caddy and Cloudflare with CI/CD gated by automated tests.

Python / AI Engineer | Growth Rise Solutions · Remote

Dec 2024 – Apr 2026

- Engineered autonomous multi-agent AI systems with CrewAI and LangChain, with custom memory modules and tool orchestration for complex, multi-step tasks.
- Designed event-driven n8n workflows connecting AI agents to CRMs, messaging services and data sources for hands-off business processes.
- Built high-concurrency FastAPI backends that turn AI logic into production APIs serving real users, plus SEO-friendly Next.js and Streamlit front ends.
- Built a hybrid (collaborative + content-based) LightFM recommendation microservice for a live grocery app (box.com.pk) to improve product discovery and engagement.
- Containerised services with Docker and automated deployments to Railway through GitHub Actions; owned versioned releases, monitoring and rollback.

Software Engineer | Bitnine Global · Islamabad*Apr 2023 – Dec 2024*

- Designed a Python test-automation framework for regression, smoke and sanity testing across Bash, PowerShell and CMD, cutting manual QA effort and raising release confidence.
- Engineered an Oracle-to-PostgreSQL migration tool in Python with both a CLI and a desktop GUI: DDL extraction, schema transformation and end-to-end data migration.
- Architected the migration tool for extensibility so additional databases (DB2, MySQL) plug in without rewriting core logic.

Full-Stack Developer (MEAN) | Zepto Systems Ltd. · Islamabad*Apr 2021 – Aug 2022*

- Built dynamic, responsive interfaces with Angular and AngularJS for a full-stack web application.
- Designed Express.js REST APIs on MySQL and integrated Strapi as a headless CMS to speed up back-end development.
- Used Pug templating for clean, maintainable views, improving velocity on UI changes.

PROJECTS

Smart Cold Storage Dashboard: led full-stack development of a cold-storage automation dashboard with real-time temperature monitoring, plus ML models that forecast temperature trends and recommend optimal storage conditions for predictive maintenance and energy efficiency.

Kids Learning Platform: web platform pairing an eye-tracking model for real-time engagement analysis with APIs around a fine-tuned GPT-3 model that personalises learning content; owned front end and back end.

Case studies: muzzammil242.github.io/#case-studies (architecture write-ups across platform, AI and data work)

EDUCATION

BS Computer Science | National University of Sciences and Technology (NUST), SEECS, Islamabad*2019 – 2023*

Coursework: Artificial Intelligence, NLP, Distributed Computing, Digital Image Processing, Database Systems, Operating Systems